HARBOR MEDICAL + OFFICE CENTRE

Benchmark B02 | Two floors | Irregular L footprint | Revision 1

Purpose

Test independent PDF quantity takeoff, editable BOQ, controlled costing, source-linked 3D and room-specific interior proposals.

Geometry

Two dimensioned vector floor plans. 28 x 24 m maximum envelope with an 8 x 6 m northeast notch. Four main X bays and four Y bands. Upper atrium void and a separate stair-zone deduction.

Scope

Internal fit-out only: masonry, glass partitions, plaster, wall/floor finishes, ceilings, openings and the explicitly declared MEP services. No foundations, frame, suspended slabs, roof waterproofing, external works or medical equipment.

Measurement

Clear-net floor areas and room finished faces. Full grid CL masonry lengths; no T-junction overlap deduction. Door and window instances are unique across the project. Do not count symbols a second time.

3D expectations

Use original plan coordinates and local floor heights. Display Ground and First separately. Keep the upper void and stair zone empty. Furniture is a proposed visual arrangement and must not enter measured BOQ.

Price control

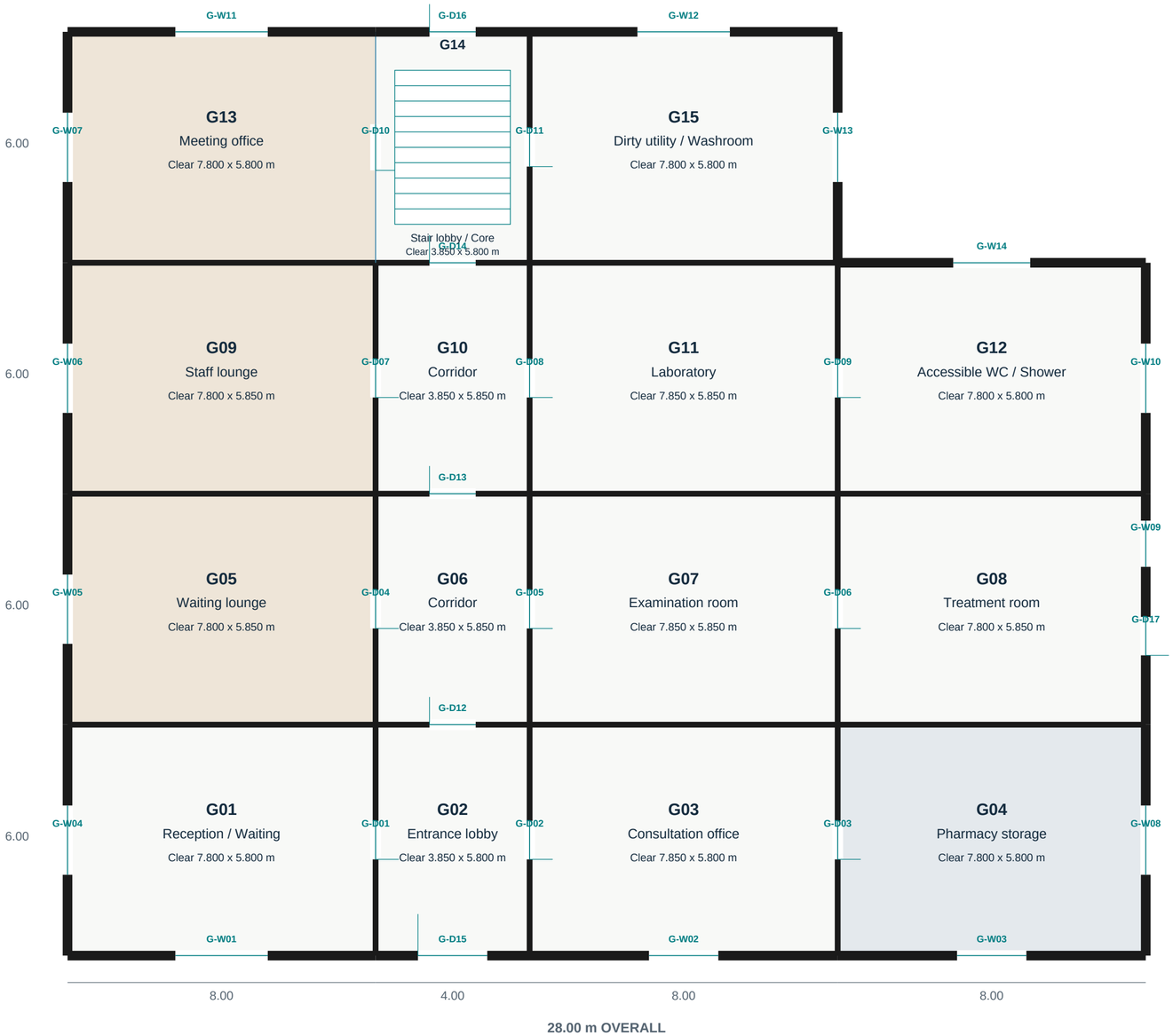
A separate CONTROL PDF contains the answer ledger. Do not upload it for analysis. Prices in this INPUT are fictional benchmark overrides, not current UAE market rates. Compare quantities first if your catalog uses different rates.

Sheet	Contents		
A101	Ground architectural plan	A402	First door instances
A102	First architectural plan	A403	Window instances - both floors
A201	Sections and measurement conventions	M101	MEP declared scope and route measurement basis
A301	Room finish register	M102	Electrical, mechanical and sanitary counts
A302	Finish legends and surface measurement rules	Q101	Fictional test unit rates
A401	Ground door instances	Q102	MEP test unit rates and quotation rules

A101 / GROUND FLOOR / ARCHITECTURAL PLAN

Vector master | 1:100 at original A2 | Coordinates in metres

Units: metres unless noted.
Scale 1:100 at original sheet.
CL polygon: (0,0) (28,0) (28,18) (20,18) (20,24) (0,24).
Grid X: 0 / 8 / 12 / 20 / 28. Grid Y: 0 / 6 / 12 / 18 / 24.
Solid external wall E01 = 250 mm CMU. Solid internal wall P01 = 150 mm CMU.
GL01: full-height glass on the complete G13-to-G14.
Stair zone: X8.5-11.5 / Y19-23. Exclude stair footprint from floor and ceiling.



LEVELS

Ground FFL +0.000. First FFL +4.200. Ground local wall height 3.800 m; first local wall height 3.300 m.

SOURCE ROOM USE

Room labels govern proposed furniture use. No furniture symbols are included in quantity takeoff.

UPPER VOID

F01 has no floor or ceiling. G01 ceiling is the single roof soffit above this atrium. Upper enclosure walls still have source wall finishes.

CONTROL

Opening ID tags refer to A401-A403. Solid partitions are measured on grid CL. Glass partitions are separate from masonry.

DRAWING LIMITATION

Stair treads and proposed furnishings are visual only. No stair structure, guards or clinical equipment are designed in this benchmark.

A102 / FIRST FLOOR / ARCHITECTURAL PLAN

Vector master | 1:100 at original A2 | Coordinates in metres

Units: metres unless noted.
Scale 1:100 at original sheet.
CL polygon: (0,0) (28,0) (28,18) (20,18) (20,24) (0,24).
Grid X: 0 / 8 / 12 / 20 / 28. Grid Y: 0 / 6 / 12 / 18 / 24.
Solid external wall E01 = 250 mm CMU. Solid internal wall P01 = 150 mm CMU.
GL01: full-height glass on the complete F13-to-F14.
Stair zone: X8.5-11.5 / Y19-23. Exclude stair opening from floor and ceiling.

LEVELS

Ground FFL +0.000. First FFL +4.200. Ground local wall height 3.800 m; first local wall height 3.300 m.

SOURCE ROOM USE

Room labels govern proposed furniture use. No furniture symbols are included in quantity takeoff.

UPPER VOID

F01 has no floor or ceiling. G01 ceiling is the single roof soffit above this atrium. Upper enclosure walls still have source wall finishes.

CONTROL

Opening ID tags refer to A401-A403. Solid partitions are measured on grid CL. Glass partitions are separate from masonry.

DRAWING LIMITATION

Stair treads and proposed furnishings are visual only. No stair structure, guards or clinical equipment are designed in this benchmark.

A201 / SECTIONS + MEASUREMENT CONVENTIONS

Controlling dimensions and explicit scope limits

GROUND external and internal walls 3.800 m above ground FFL. FIRST external and internal walls 3.300 m above first FFL.

Ground FFL +0.000 m; first FFL +4.200 m; upper wall head +7.500 m. G01 exposed roof-soffit ceiling is at +7.500 m and belongs to G01 only.

Room clear faces offset 125 mm from external CL; 75 mm from internal CL.

Partition lengths use full grid CL lengths. Do not deduct T-junction overlaps; deduct every scheduled door/window opening, regardless of size.

For each room, measure 2 x (clear width + clear depth) x local wall height. Subtract the entire clear length of any full-height GL01 face.

PL01: All solid clear room wall faces, less openings; includes under wet wall tiles. PA01: PL01 net faces minus WT01 tiled faces.

G12/G15/F12: all four clear walls to 2.400 m; less doors/windows below 2.400. Wet rooms receive full clear floor waterproofing.

WP02: 0.200 m high x wet-room clear perimeter minus door widths. No additional skirting in wet rooms.

Ceiling areas are net clear room areas, with the same core stair-zone deductions as floors.

F01 is VOID - no floor / no ceiling. Do not count its ceiling a second time. Wall finishes on the upper atrium enclosure remain included.

All other dry rooms: skirting uses clear perimeter minus door widths and full-height glass clear faces. Window widths are not deducted from skirting.

No deductions for finishes thickness. No allowances for unshown risers, concealed pipe lengths, wastage or structural elements. The MEP schedule is the sole declared MEP takeoff basis.

A301 / ROOM FINISH REGISTER

Unique physical rooms; VOID is deliberately excluded from floor and ceiling

GF room	Floor	Ceiling	FF room	Floor	Ceiling
G01	FL01	C03	F01	VOID	VOID
G02	FL01	C01	F02	FL01	C01
G03	FL01	C01	F03	FL01	C01
G04	FL03	C02	F04	FL03	C02
G05	FL02	C01	F05	FL01	C01
G06	FL01	C01	F06	FL01	C01
G07	FL01	C01	F07	FL01	C01
G08	FL01	C01	F08	FL02	C01
G09	FL02	C01	F09	FL01	C01
G10	FL01	C01	F10	FL01	C01
G11	FL01	C01	F11	FL01	C01
G12	FL01	C02	F12	FL01	C02
G13	FL02	C01	F13	FL02	C01
G14	FL01	C01	F14	FL01	C01
G15	FL01	C02	F15	FL03	C02

FL01 porcelain; FL02 engineered wood; FL03 epoxy. Full legends and inclusions on A302.

A302 / FINISH LEGENDS + SURFACE RULES

Rates and quantities must follow these material scopes

Code	Floor finish	Skirting rule
FL01	Porcelain tile	Dry rooms: matching skirting; wet rooms: none
FL02	Engineered wood	Dry rooms: timber skirting
FL03	Epoxy resin	Dry rooms: resin skirting

Code	Description	Measurement rule
C01	Gypsum board ceiling	Paint; board rate excludes paint
C02	Moisture-resistant gypsum ceiling	Paint; board rate excludes paint
C03	Exposed concrete soffit	Paint only; no gypsum board

Code	Material	Measurement rule
GL01	Full-height fixed internal glass	Net CL length x local height less D04 glass door openings; BOQ m2

All fixed-glass areas exclude glass door leaves and frames, which are priced separately as D04 door sets. No external weather paint is included.

A401 / DOOR SCHEDULE / UNIQUE INSTANCES

Ground source tags and host endpoints; no repeated type counts

Instance	Type	W m	H m	Spaces	CL endpoints (x,y) m
G-D01	D01	1.000	2.200	G01 / G02	(8,2.5) to (8,3.5)
G-D02	D01	1.000	2.200	G02 / G03	(12,2.5) to (12,3.5)
G-D03	D01	1.000	2.200	G03 / G04	(20,2.5) to (20,3.5)
G-D04	D01	1.000	2.200	G05 / G06	(8,8.5) to (8,9.5)
G-D05	D01	1.000	2.200	G06 / G07	(12,8.5) to (12,9.5)
G-D06	D01	1.000	2.200	G07 / G08	(20,8.5) to (20,9.5)
G-D07	D01	1.000	2.200	G09 / G10	(8,14.5) to (8,15.5)
G-D08	D01	1.000	2.200	G10 / G11	(12,14.5) to (12,15.5)
G-D09	D01	1.000	2.200	G11 / G12	(20,14.5) to (20,15.5)
G-D10	D04	1.200	2.400	G13 / G14	(8,20.4) to (8,21.6)
G-D11	D01	1.000	2.200	G14 / G15	(12,20.5) to (12,21.5)
G-D12	D01	1.200	2.200	G02 / G06	(9.4,6) to (10.6,6)
G-D13	D01	1.200	2.200	G06 / G10	(9.4,12) to (10.6,12)
G-D14	D01	1.200	2.200	G10 / G14	(9.4,18) to (10.6,18)
G-D15	D02	1.800	2.400	G02 / EXT	(9.1,0) to (10.9,0)
G-D16	D03	1.200	2.200	G14 / EXT	(9.4,24) to (10.6,24)
G-D17	D03	1.000	2.200	G08 / EXT	(28,7.8) to (28,8.8)

D01 solid internal set; D02 double entrance set; D03 external service/exit set; D04 framed glass set. Door swing/hinge not scheduled: 3D may use a clearly proposed closed leaf.

A402 / DOOR SCHEDULE / UNIQUE INSTANCES

First source tags and host endpoints; no repeated type counts

Instance	Type	W m	H m	Spaces	CL endpoints (x,y) m
F-D01	D01	1.000	2.200	F02 / F03	(12,2.5) to (12,3.5)
F-D02	D01	1.000	2.200	F03 / F04	(20,2.5) to (20,3.5)
F-D03	D01	1.000	2.200	F05 / F06	(8,8.5) to (8,9.5)
F-D04	D01	1.000	2.200	F06 / F07	(12,8.5) to (12,9.5)
F-D05	D01	1.000	2.200	F07 / F08	(20,8.5) to (20,9.5)
F-D06	D01	1.000	2.200	F09 / F10	(8,14.5) to (8,15.5)
F-D07	D01	1.000	2.200	F10 / F11	(12,14.5) to (12,15.5)
F-D08	D01	1.000	2.200	F11 / F12	(20,14.5) to (20,15.5)
F-D09	D04	1.200	2.400	F13 / F14	(8,20.4) to (8,21.6)
F-D10	D01	1.000	2.200	F14 / F15	(12,20.5) to (12,21.5)
F-D11	D01	1.200	2.200	F02 / F06	(9.4,6) to (10.6,6)
F-D12	D01	1.200	2.200	F06 / F10	(9.4,12) to (10.6,12)
F-D13	D01	1.200	2.200	F10 / F14	(9.4,18) to (10.6,18)

D01 solid internal set; D02 double entrance set; D03 external service/exit set; D04 framed glass set. Door swing/hinge not scheduled: 3D may use a clearly proposed closed leaf.

A403 / WINDOW SCHEDULE / UNIQUE INSTANCES

Ground + First | Every tag is one physical window | No repeated symbols

Instance	Type	W m	H m	Sill m	Room	CL endpoints (x,y) m
G-W01	W01	2.400	1.500	0.900	G01	(2.8,0) to (5.2,0)
G-W02	W01	1.800	1.500	0.900	G03	(15.1,0) to (16.9,0)
G-W03	W01	1.800	1.500	0.900	G04	(23.1,0) to (24.9,0)
G-W04	W01	1.800	1.500	0.900	G01	(0,2.1) to (0,3.9)
G-W05	W01	1.800	1.500	0.900	G05	(0,8.1) to (0,9.9)
G-W06	W01	1.800	1.500	0.900	G09	(0,14.1) to (0,15.9)
G-W07	W01	1.800	1.500	0.900	G13	(0,20.1) to (0,21.9)
G-W08	W01	1.800	1.500	0.900	G04	(28,2.1) to (28,3.9)
G-W09	W01	1.200	1.500	0.900	G08	(28,10.1) to (28,11.3)
G-W10	W01	1.800	1.500	0.900	G12	(28,14.1) to (28,15.9)
G-W11	W01	2.400	1.500	0.900	G13	(2.8,24) to (5.2,24)
G-W12	W01	2.400	1.500	0.900	G15	(14.8,24) to (17.2,24)
G-W13	W01	1.800	1.500	0.900	G15	(20,20.1) to (20,21.9)
G-W14	W01	2.000	1.500	0.900	G12	(23,18) to (25,18)
F-W01	W01	2.400	1.500	0.900	F01	(2.8,0) to (5.2,0)
F-W02	W01	1.800	1.500	0.900	F03	(15.1,0) to (16.9,0)
F-W03	W01	1.800	1.500	0.900	F04	(23.1,0) to (24.9,0)
F-W04	W01	1.800	1.500	0.900	F01	(0,2.1) to (0,3.9)
F-W05	W01	1.800	1.500	0.900	F05	(0,8.1) to (0,9.9)
F-W06	W01	1.800	1.500	0.900	F09	(0,14.1) to (0,15.9)
F-W07	W01	1.800	1.500	0.900	F13	(0,20.1) to (0,21.9)
F-W08	W01	1.800	1.500	0.900	F04	(28,2.1) to (28,3.9)
F-W09	W01	1.200	1.500	0.900	F08	(28,10.1) to (28,11.3)
F-W10	W01	1.800	1.500	0.900	F12	(28,14.1) to (28,15.9)
F-W11	W01	2.400	1.500	0.900	F13	(2.8,24) to (5.2,24)
F-W12	W01	2.400	1.500	0.900	F15	(14.8,24) to (17.2,24)
F-W13	W01	1.800	1.500	0.900	F15	(20,20.1) to (20,21.9)
F-W14	W01	2.000	1.500	0.900	F12	(23,18) to (25,18)
F-W15	W01	1.800	1.500	0.900	F02	(9.1,0) to (10.9,0)
F-W16	W01	1.200	1.500	0.900	F14	(9.4,24) to (10.6,24)

M101 / DECLARED MEP MEASUREMENT BASIS

Services are explicit source quantities; do not infer routing from furniture or architecture

Total lengths below include all horizontal runs and the stated project risers once. No quantity is repeated per floor. Pipe lengths are centreline lengths; duct quantity is developed sheet area, not linear metres. Insulation quantity equals the included CHW supply + return length.

Code	Source operation	Quantity	Unit
CW20	Cold water pipe 20 mm	58	m
CW25	Cold water pipe 25 mm	32	m
HW20	Hot water pipe 20 mm	26	m
DW50	Waste pipe 50 mm	42	m
DW110	Soil pipe 110 mm	28	m
CHW25	FCU supply + return pipe 25 mm	176	m
INS25	Pipe insulation 25 mm	176	m
DU01	Galvanized duct 0.8 mm - developed sheet area	92	m2
CB20	Power cable 3 x 2.5 mm2	420	m
CB40	Power cable 3 x 4 mm2	186	m
CT01	Cable tray 150 x 50 mm	96	m

CW20 = 31 m GF + 23 m FF + 4 m project riser. CW25 = 16 + 12 + 4 m. HW20 = 13 + 9 + 4 m. DW50 = 23 + 15 + 4 m. DW110 = 14 + 10 + 4 m.

CHW25 = 92 m GF + 76 m FF + 8 m paired project risers. INS25 covers the same 176 m once. CB20 = 220 + 200 m; CB40 = 96 + 90 m; CT01 = 50 + 46 m.

M102 / EQUIPMENT + POINT SCHEDULE

Project counts, not counts per floor; each device is included once

Code	Description	Ground	First	Project qty
LT01	LED lights complete	63	63	126
SO01	13A socket outlets complete	57	57	114
FCU01	Fan coil units - 3.5 kW	7	8	15
DB01	Distribution boards	2	2	4
SD01	Smoke detectors	19	19	38
WC01	Toilet pans	2	1	3
WB01	Wash basins	3	1	4
SH01	Shower sets	1	1	2
SK01	Clinical sink / kitchen sink	2	1	3

Sanitary host allocation agrees with the GF/FF summary: WC pans G12/G15/F12, basins G12 x2 + G15 + F12, shower sets G12/F12, sinks G07/G11/F11. Count these host identities once.

Mechanical and electrical rates include declared accessories only. Do not invent medical equipment, fire suppression, ICT/security devices or equipment from suggested 3D furniture.

Q101 / FICTIONAL TEST UNIT RATES

Controlled benchmark only | AED | Material + labour | These are not market quotations

Code	Operation	Unit	AED/unit
MS01	250 mm external masonry net	m2	125.00
MS02	150 mm internal masonry net	m2	98.00
GL01	Fixed internal glass net of glass doors	m2	650.00
PL01	Internal plaster incl. under wet tiles	m2	32.00
PA01	Internal wall paint excluding wall tiles	m2	16.00
WT01	Wet room wall tile to 2.4 m	m2	85.00
WP01	Wet floor waterproofing	m2	55.00
WP02	Waterproof upstand 0.2 m	m2	45.00
FL01	Porcelain floor tile	m2	120.00
FL02	Engineered wood floor	m2	155.00
FL03	Epoxy floor	m2	95.00
C01	Gypsum ceiling excluding paint	m2	90.00
CP01	Ceiling paint incl. exposed soffit	m2	16.00
SKT01	Dry-room skirting	m	22.00
W01	External windows	m2	900.00
D01	Door set D01	nr	1100.00
D02	Door set D02	nr	2800.00
D03	Door set D03	nr	1450.00
D04	Door set D04	nr	2200.00

No project expected quantities or expected total are supplied here. Quantity takeoff must be obtained from A101-A403 and M101-M102. Use these rates only for a controlled-price comparison.

Q102 / MEP TEST RATES + QUOTATION RULES

Controlled benchmark only | AED | Separate answer ledger is not part of this INPUT

Code	Operation	Unit	AED/unit
CW20	Cold water pipe 20 mm	m	38.00
CW25	Cold water pipe 25 mm	m	48.00
HW20	Hot water pipe 20 mm	m	44.00
DW50	Waste pipe 50 mm	m	52.00
DW110	Soil pipe 110 mm	m	78.00
CHW25	FCU supply + return pipe 25 mm	m	58.00
INS25	Pipe insulation 25 mm	m	22.00
DU01	Galvanized duct 0.8 mm - developed sheet area	m2	105.00
CB20	Power cable 3 x 2.5 mm2	m	12.00
CB40	Power cable 3 x 4 mm2	m	18.00
CT01	Cable tray 150 x 50 mm	m	65.00
LT01	LED lights complete	nr	145.00
SO01	13A socket outlets complete	nr	90.00
FCU01	Fan coil units - 3.5 kW	nr	2100.00
DB01	Distribution boards	nr	1450.00
SD01	Smoke detectors	nr	85.00
WC01	Toilet pans	nr	780.00
WB01	Wash basins	nr	620.00
SH01	Shower sets	nr	560.00
SK01	Clinical sink / kitchen sink	nr	720.00

Cost protocol: round each quantity x rate line to AED 0.01, then sum the measured direct-cost lines. Add 10% overhead/profit once; then add 5% test VAT to direct cost + overhead/profit. Do not add an extra contingency or wastage allowance.